Introduction to Artificial Intelligence and Machine Learning (CSN11601)

Assignment-3

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Q1. Write a python program to find largest of three numbers.

A1. CODE:

```
a=int(input('enter no. 1 : '))
b=int(input('enter no. 2 : '))
c=int(input('enter no. 3 : '))
if a>b:
   if a>c:
      print('the greatest no. is',a)
   else:
      print('the greatest no. is',c)
else:
  if b>c :
    print('the greatest no. is',b)
  else :
     print('the greatest no. is',c)
```

OUTPUT:

```
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/AI-ML/Assignment-3$ python3 p1.py
enter no. 1 : 12
enter no. 2 : 14
enter no. 3 : 16
the greatest no. is 16
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/AI-ML/Assignment-3$ python3 p1.py
enter no. 1 : 14
enter no. 2 : 12
enter no. 3 : 10
the greatest no. is 14
```

Q2. Read two integer numbers from the keyboard and print minimum value using ternary operator.

A2. CODE:

```
a = int(input("Enter first integer: "))
b = int(input("Enter second integer: "))
minimum=a if a<b else b
print("Minimum value is:",minimum)
```

OUTPUT:

```
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/AI-ML/Assignment-3$ python3 p2.py
Enter first integer: 67
Enter second integer: 889
Minimum value is: 67
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/AI-ML/Assignment-3$ python3 p2.py
Enter first integer: 12
Enter second integer: 14
Minimum value is: 12
```

Q3. Write a Python program to compute the GCD of two numbers.
A3. CODE:

```
a=int(input("Enter first number: "))
b=int(input("Enter second number: "))
gcd=1
for i in range(1,min(a,b)+1):
    if a%i==0 and b%i== 0:
        gcd=i
print("GCD of",a,"and",b,"is:",gcd)
```

OUTPUT:

```
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/AI-ML/Assignment-3$ python3 p3.py
Enter first number: 7
Enter second number: 8
GCD of 7 and 8 is: 1
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/AI-ML/Assignment-3$ python3 p3.py
Enter first number: 16
Enter second number: 20
GCD of 16 and 20 is: 4
```

Q4. Write a python program to that accepts length of three sides of a triangle as inputs. The program should indicate whether or not the triangle is a right-angled triangle (use Pythagorean theorem).
A4. CODE:

```
a = int(input("Enter first side: "))
b = int(input("Enter second side: "))
c = int(input("Enter third side: "))
sides = sorted([a,b,c])
if sides[0]**2+sides[1]**2==sides[2]**2:
    print("It is a right-angled triangle.")
else:
    print("It is NOT a right-angled triangle.")
```

OUTPUT:

```
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/AI-ML/Assignment-3$ python3 p4.py
Enter first side: 3
Enter second side: 5
Enter third side: 4
It is a right-angled triangle.
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/AI-ML/Assignment-3$ python3 p4.py
Enter first side: 12
Enter second side: 13
Enter third side: 14
It is NOT a right-angled triangle.
```

Q5. Write a program to print a given number is Armstrong or not.
A5. CODE:

```
num = int(input("Enter a number: "))
digits = str(num)
power = len(digits)
sum_of_powers = sum(int(digit)**power for digit in digits)
if sum_of_powers==num:
    print(num,"is an Armstrong number.")
else:
    print(num,"is NOT an Armstrong number.")
```

OUTPUT:

```
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/AI-ML/Assignment-3$ python3 p5.py
Enter a number: 16
16 is NOT an Armstrong number.
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/AI-ML/Assignment-3$ python3 p5.py
Enter a number: 153
153 is an Armstrong number.
```

Q6. Write a program to print first 100 elements of a Fibonacci series
A6. CODE:

```
n1,n2=0,1
count=0
print("First 20 Fibonacci numbers:")
while count<20:
    print(n1,end=" ")
    n1,n2=n2,n1+n2
    count+=1
print()
```

OUTPUT:

```
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/AI-ML/Assignment-3$ python3 p6.py
First 20 Fibonacci numbers:
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584 4181
```